Chapter 4, Section 6

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1. A rational curve of degree 4 in $\mathbb{P}^3$ is contained in a unique quadric surface Q, and Q is necessarily nonsingular.

Proof. Let X be a rational curve of degree 4 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now the middle vector space has dimension 10, and $\mathcal{O}_X(2)$ corresponds to a linear system of degree 8, so it has dimension at most 9 by Riemann-Roch. Then, $\mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2))$ has dimension at least 1, so X is contained in a quadric surface Q. If X is contained in another quadric surface, then it must be a complete intersection by reason of its degree, which implies $\omega_X = \mathcal{O}_X$ (II, Ex. 8.4), a contradiction. If Q is singular, then X must have genus 1 (6.4.1), which is a contradiction. Hence, Q must be a nonsingular quadric surface. □

2. A rational curve of degree 5 in $\mathbb{P}^3$ is always contained in a cubic surface, but there are such curves which are not contained in any quadric surface.

Proof. Let X be a rational curve of degree 5 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(3)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(3)) \longrightarrow \cdots.$$

Now, the middle vector space has dimension 20, and the vector space on the right has dimension 16 by Riemann-Roch on X (note that $\mathcal{O}_X(3)$ corresponds to a divisor of degree 15). So we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(3)) \geq 4$.

A degree 5 curve in $\mathbb{P}^3$ contained in a quadric cone must have genus 2 (6.4.1), so if X is contained in a quadric surface, then it must be a nonsingular quadric surface. Choosing a degree 5 embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$ is the same as picking a 4-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(5))$, which has dimension 6. Thus, the family of embeddings $\mathbb{P}^1 \to \mathbb{P}^3$ of degree 5 is parametrized by the Grassmannian $\mathrm{Gr}(4, 6)$, which has dimension $4(6-4) = 8$. On the other hand, if X is contained in a nonsingular quadric surface $Q \subset \mathbb{P}^3$, then it corresponds to a divisor of type $(1, 4)$ on Q. So choosing an embedding of $\mathbb{P}^1$ as a curve of type $(1, 4)$ in Q is the same as picking a 2-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1)$ and a 2-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(4))$. Thus, the family of embeddings $\mathbb{P}^1 \to Q$ is parametrized by $\mathrm{Gr}(2, 2) \times \mathrm{Gr}(2, 5)$, which has dimension $1(2-1) + 2(5-2) = 7$. Since the family of embeddings $Q \to \mathbb{P}^3$ has dimension one, there are plenty of embeddings of $\mathbb{P}^1$ as a degree 5 curve in $\mathbb{P}^3$ which are not contained in any quadric surface. □

3. A curve of degree 5 and genus 2 in $\mathbb{P}^3$ is contained in a unique quadric surface Q. Show that for any abstract curve X of genus 2, there exist embeddings of degree 5 in $\mathbb{P}^3$ for which Q is nonsingular, and there exist other embeddings of degree 5 for which Q is singular.

Proof. Let X be a curve of degree 5 and genus 2 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

The middle vector space has dimension 10, and the vector space on the right has dimension 9 by Riemann-Roch on X (note that $\mathcal{O}_X(2)$ corresponds to a nonspecial divisor of degree 10). So we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(2)) \geq 1$. By reason of degree, X cannot be contained in two distinct quadric surfaces, so there is a unique irreducible quadric surface Q containing X.

Similar to (Ex. 6.2), if X is a genus 2 curve of degree 5 in a nonsingular quadric surface $Q \subset \mathbb{P}^3$, then it corresponds to a divisor of type $(2, 3)$ on Q. Thus, choosing an embedding of X as a curve of type $(2, 3)$ in $Q \subset \mathbb{P}^3$ is the same as picking a g^1_2 and g^1_3 on X. Since X is hyperelliptic, it always has a g^1_2 and g^1_3, so there are plenty of embeddings of X into a nonsingular quadric surface. In fact, any g^1_3 on X is obtained by adding a point to the unique g^1_2, so the family of g^1_3's on X has dimension one, implying that the family of embeddings of X into a nonsingular quadric has dimension one. On the other hand, any effective divisor of degree 5 gives an embedding of X in $\mathbb{P}^3$ (Ex. 3.1), which is the same as picking 5 points in X, so the family of embeddings $X \to \mathbb{P}^3$ has dimension at least 5.

□

4. There is no curve of degree 9 and genus 11 in $\mathbb{P}^3$.

Proof. Suppose there exists a curve $X \subseteq \mathbb{P}^3$ of degree 9 and genus 11, and let $\mathcal{I}$ be its ideal sheaf. We have an exact sequence

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Then $\mathcal{O}_X(2)$ is a very ample line bundle corresponding to a linear system of degree 18, say $|\mathcal{O}_X(2)| = |D|$ for some very ample divisor D of degree 18. Riemann-Roch gives

$$\dim |D| - \dim |K - D| = 8.$$

If D is nonspecial, then $\dim |D| = 8$. If D is special, Clifford's theorem (5.4) says

$$\dim |D| \leq \frac{1}{2} \deg D = 9,$$

and we have an equality if and only if X is hyperelliptic and D is a multiple of the unique g^1_2 on X. But D is very ample, so it cannot be a multiple of the g^1_2, and the inequality must be strict. We conclude that

$$h^0(X, \mathcal{O}_X(2)) = 1 + \dim |D| \leq 9,$$

and therefore, $\mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \neq 0$. Hence, X must be contained in a quadric surface.

Following the discussion in (6.4.1), if X is contained in nonsingular quadric surface, then it corresponds to a divisor of type (a, b) such that

$$a + b = 9, \quad ab - a - b + 1 = 11,$$

which is not possible. If X is contained in a singular quadric surface, then X must have genus 12. Hence, there is no curve of degree 9 and genus 11 in $\mathbb{P}^3$. □

5. If X is a complete intersection of surfaces of degree a, b in $\mathbb{P}^3$, then X does not lie on any surface of degree $< \min(a, b)$.

Proof. If X is a complete intersection of surfaces H, H' of degree a, b respectively, in $\mathbb{P}^3$, then X has degree ab. Without loss of generality, suppose $a \leq b$. If H'' is surface of degree $c < a$ containing X, then the irreducible components of $H \cap H''$ can have degree at most ac (I, 7.7), which is strictly less than ab, so X cannot be an irreducible component of $H \cap H''$. Hence, X cannot lie on any surface of degree $< a$. □